

Problem 1: Online Shopping System

Scenario

An e-commerce company wants to maintain customer information and their product orders.

Students need to:

- Create the tables.
- Insert the following records manually.
- Practice UPDATE, DELETE and JOIN operations.

Table 1: Customers

Customer ID	Customer Name	City	Phone
101	Amit Sharma	Delhi	9876543201
102	Priya Verma	Mumbai	9876543202
103	Rahul Singh	Pune	9876543203
104	Sneha Gupta	Jaipur	9876543204
105	Arjun Patel	Ahmedabad	9876543205
106	Neha Joshi	Surat	9876543206
107	Karan Mehta	Chandigarh	9876543207
108	Pooja Nair	Kochi	9876543208
109	Rohan Das	Kolkata	9876543209
110	Anjali Roy	Lucknow	9876543210
111	Vivek Kumar	Patna	9876543211
112	Meera Shah	Indore	9876543212
113	Deepak Yadav	Nagpur	9876543213
114	Nisha Jain	Bhopal	9876543214
115	Mohit Saini	Noida	9876543215
116	Aisha Khan	Hyderabad	9876543216
117	Saurabh Mishra	Kanpur	9876543217
118	Ritu Agarwal	Agra	9876543218
119	Akash Malhotra	Ludhiana	9876543219
120	Simran Kaur	Amritsar	9876543220
121	Harsh Vyas	Udaipur	9876543221
122	Divya Rao	Mysore	9876543222
123	Manish Saxena	Dehradun	9876543223
124	Kavita Singh	Ranchi	9876543224
125	Aditya Bose	Bhubaneswar	9876543225

Table 2: Orders

Order ID	Customer ID	Product Name	Quantity	Price (₹)	Order Date
1001	101	Laptop	1	65000	2026-01-10
1002	102	Smartphone	2	25000	2026-01-11
1003	103	Headphones	3	2500	2026-01-12
1004	104	Keyboard	2	1800	2026-01-12
1005	105	Mouse	4	700	2026-01-13
1006	106	Monitor	1	15000	2026-01-14
1007	107	Printer	1	12000	2026-01-15
1008	108	Tablet	2	28000	2026-01-16
1009	109	Webcam	3	2200	2026-01-17

1010	110	SSD	2	5000	2026-01-18
1011	111	Laptop	1	72000	2026-01-19
1012	112	Smartphone	1	30000	2026-01-20
1013	113	Smart Watch	2	8500	2026-01-21
1014	114	Bluetooth Speaker	2	3500	2026-01-22
1015	115	Router	1	2800	2026-01-23
1016	116	Power Bank	3	1800	2026-01-24
1017	117	Pendrive	5	900	2026-01-25
1018	118	External HDD	2	6500	2026-01-26
1019	119	Gaming Mouse	1	3200	2026-01-27
1020	120	Mechanical Keyboard	1	5500	2026-01-28
1021	121	Laptop Bag	2	1800	2026-01-29
1022	122	UPS	1	6500	2026-01-30
1023	123	Graphics Card	1	42000	2026-01-31
1024	124	CPU Cabinet	1	4800	2026-02-01
1025	125	Projector	1	35000	2026-02-02

Practice Questions

CREATE

- Create a database named **OnlineShoppingDB**.
- Create the **Customers** table.
- Create the **Orders** table with an appropriate foreign key.

INSERT

- Insert all 25 customer records.
- Insert all 25 order records.

UPDATE

- Change the city of **Customer ID 115** from **Noida** to **Gurugram**.
- Update the phone number of **Customer ID 122**.
- Change the quantity of **Order ID 1017** from **5** to **8**.
- Increase the price of the **Projector** to **₹37,000**.
- Change the product name **Router** to **WiFi Router**.

DELETE

- Delete **Customer ID 125**.
- Delete **Order ID 1025**.
- Delete the order for **Gaming Mouse**.
- Delete the customer whose name is **Harsh Vyas**.
- Delete the order placed on **2026-01-14**.

JOIN

- Display customer name along with the product they ordered.
- Display customer city along with product name.
- Display order ID, customer name, product name, and order date.
- Display all orders along with customer phone numbers.
- Display customers who have placed orders for **Laptop**.

Problem 2: Hospital Management System

Scenario

A multi-specialty hospital wants to maintain information about its doctors and patients. Each patient is assigned to one doctor. Students need to create the tables, insert the records, and practice **CREATE**, **INSERT**, **UPDATE**, **DELETE**, and **JOIN** operations.

Table 1: Doctors

Doctor ID	Doctor Name	Specialization	Experience (Years)
201	Dr. Amit Sharma	Cardiologist	15
202	Dr. Priya Verma	Neurologist	12
203	Dr. Rahul Singh	Orthopedic	10
204	Dr. Sneha Gupta	Pediatrician	8
205	Dr. Arjun Patel	Dermatologist	9
206	Dr. Neha Joshi	Gynecologist	14
207	Dr. Karan Mehta	ENT Specialist	11
208	Dr. Pooja Nair	Ophthalmologist	7
209	Dr. Rohan Das	Psychiatrist	13
210	Dr. Anjali Roy	General Physician	16
211	Dr. Vivek Kumar	Urologist	9
212	Dr. Meera Shah	Oncologist	17
213	Dr. Deepak Yadav	Nephrologist	12
214	Dr. Nisha Jain	Pulmonologist	10
215	Dr. Mohit Saini	Endocrinologist	8
216	Dr. Aisha Khan	Gastroenterologist	11
217	Dr. Saurabh Mishra	Rheumatologist	14
218	Dr. Ritu Agarwal	Surgeon	18
219	Dr. Akash Malhotra	Cardiologist	6
220	Dr. Simran Kaur	Pediatrician	5
221	Dr. Harsh Vyas	Dentist	9
222	Dr. Divya Rao	Neurologist	13
223	Dr. Manish Saxena	Orthopedic	7
224	Dr. Kavita Singh	General Physician	15
225	Dr. Aditya Bose	Dermatologist	8

Table 2: Patients

Patient ID	Patient Name	Doctor ID	Disease	Admission Date
301	Aarav Gupta	201	Heart Disease	2026-02-01
302	Siya Sharma	202	Migraine	2026-02-02
303	Rohan Verma	203	Fracture	2026-02-03
304	Ananya Singh	204	Viral Fever	2026-02-04
305	Kabir Patel	205	Skin Allergy	2026-02-05
306	Ishita Joshi	206	Pregnancy Checkup	2026-02-06
307	Vihaan Mehta	207	Ear Infection	2026-02-07
308	Myra Nair	208	Eye Infection	2026-02-08
309	Arjun Das	209	Anxiety	2026-02-09
310	Kiara Roy	210	Diabetes	2026-02-10
311	Aditya Kumar	211	Kidney Stone	2026-02-11

312	Aditi Shah	212	Cancer Screening	2026-02-12
313	Dev Yadav	213	Kidney Infection	2026-02-13
314	Riya Jain	214	Asthma	2026-02-14
315	Yash Saini	215	Thyroid	2026-02-15
316	Mehak Khan	216	Ulcer	2026-02-16
317	Aryan Mishra	217	Arthritis	2026-02-17
318	Tanvi Agarwal	218	Appendicitis	2026-02-18
319	Kunal Malhotra	219	Heart Checkup	2026-02-19
320	Pari Kaur	220	Cold & Cough	2026-02-20
321	Om Vyas	221	Tooth Decay	2026-02-21
322	Diya Rao	222	Epilepsy	2026-02-22
323	Rudra Saxena	223	Knee Pain	2026-02-23
324	Saanvi Singh	224	High Fever	2026-02-24
325	Krish Bose	225	Acne	2026-02-25

Practice Questions

CREATE

- Create a database named **HospitalDB**.
- Create the **Doctors** table.
- Create the **Patients** table with an appropriate foreign key.

INSERT

- Insert all 25 doctor records.
- Insert all 25 patient records.

UPDATE

- Change the specialization of **Doctor ID 225** from **Dermatologist** to **Cosmetologist**.
- Update the experience of **Doctor ID 220** from **5 years** to **7 years**.
- Change the disease of **Patient ID 309** from **Anxiety** to **Depression**.
- Update the admission date of **Patient ID 320** to **2026-02-22**.
- Assign **Patient ID 323** to **Doctor ID 203**.

DELETE

- Delete **Patient ID 325**.
- Delete **Doctor ID 225**.
- Delete the patient whose name is **Pari Kaur**.
- Delete the doctor whose name is **Dr. Harsh Vyas**.
- Delete the patient admitted on **2026-02-06**.

JOIN

- Display patient names along with their doctor names.
- Display patient name, disease, and doctor's specialization.
- Display patient name, doctor name, and admission date.
- Display doctor experience along with the patients assigned to them.
- Display all patients treated by **Cardiologists**.

Problem 3: College Management System

Scenario

A college wants to maintain information about its departments and students. Each student belongs to one department. Students need to create the tables, insert the records, and practice **CREATE, INSERT, UPDATE, DELETE,** and **JOIN** operations.

Table 1: Departments

Department ID	Department Name	HOD Name	Building
301	Computer Science	Dr. Amit Sharma	Block A
302	Information Technology	Dr. Priya Verma	Block B
303	Electronics	Dr. Rahul Singh	Block C
304	Mechanical	Dr. Sneha Gupta	Block D
305	Civil	Dr. Arjun Patel	Block E
306	Electrical	Dr. Neha Joshi	Block F
307	Artificial Intelligence	Dr. Karan Mehta	Block G
308	Data Science	Dr. Pooja Nair	Block H
309	Biotechnology	Dr. Rohan Das	Block I
310	Chemical	Dr. Anjali Roy	Block J
311	Commerce	Dr. Vivek Kumar	Block K
312	Management	Dr. Meera Shah	Block L
313	Mathematics	Dr. Deepak Yadav	Block M
314	Physics	Dr. Nisha Jain	Block N
315	Chemistry	Dr. Mohit Saini	Block O
316	English	Dr. Aisha Khan	Block P
317	Economics	Dr. Saurabh Mishra	Block Q
318	Psychology	Dr. Ritu Agarwal	Block R
319	History	Dr. Akash Malhotra	Block S
320	Geography	Dr. Simran Kaur	Block T
321	Sociology	Dr. Harsh Vyas	Block U
322	Political Science	Dr. Divya Rao	Block V
323	Finance	Dr. Manish Saxena	Block W
324	Marketing	Dr. Kavita Singh	Block X
325	Human Resources	Dr. Aditya Bose	Block Y

Table 2: Students

Student ID	Student Name	Department ID	Semester	CGPA
401	Aarav Gupta	301	2	8.5
402	Siya Sharma	302	4	8.2
403	Rohan Verma	303	6	7.8
404	Ananya Singh	304	3	8.9
405	Kabir Patel	305	5	7.4
406	Ishita Joshi	306	1	9.1
407	Vihaan Mehta	307	7	8.7
408	Myra Nair	308	2	8.3
409	Arjun Das	309	4	7.6
410	Kiara Roy	310	8	8.8
411	Aditya Kumar	311	5	7.9

412	Aditi Shah	312	6	9.0
413	Dev Yadav	313	3	8.1
414	Riya Jain	314	2	7.5
415	Yash Saini	315	4	8.6
416	Mehak Khan	316	1	9.2
417	Aryan Mishra	317	7	8.0
418	Tanvi Agarwal	318	5	8.4
419	Kunal Malhotra	319	6	7.7
420	Pari Kaur	320	3	8.5
421	Om Vyas	321	2	7.9
422	Diya Rao	322	8	8.8
423	Rudra Saxena	323	4	8.1
424	Saanvi Singh	324	5	7.8
425	Krish Bose	325	6	9.3

Practice Questions

CREATE

- Create a database named **CollegeDB**.
- Create the **Departments** table.
- Create the **Students** table with an appropriate foreign key.

INSERT

- Insert all 25 department records.
- Insert all 25 student records.

UPDATE

- Change the **HOD** of the **Artificial Intelligence** department to **Dr. Rakesh Gupta**.
- Update the **Building** of the **Marketing** department from **Block X** to **Block Z**.
- Change the **CGPA** of **Student ID 417** from **8.0** to **8.6**.
- Update the **Semester** of **Student ID 406** from **1** to **2**.
- Transfer **Student ID 424** from the **Marketing** department to the **Finance** department.

DELETE

- Delete **Student ID 425**.
- Delete the **Human Resources** department.
- Delete the student named **Pari Kaur**.
- Delete the department named **History**.
- Delete the student with **Student ID 411**.

JOIN

- Display the **student name** along with their **department name**.
- Display the **student name**, **department name**, and **HOD name**.
- Display the **student name**, **semester**, and **building** of their department.
- Display all students who belong to the **Computer Science** department.
- Display the **student name**, **department name**, and **CGPA**.

Problem 4: Movie Ticket Booking System

Scenario

A multiplex cinema wants to maintain information about the movies currently running and the tickets booked by customers. Each booking is associated with one movie. Students need to create the tables, insert the records, and practice **CREATE**, **INSERT**, **UPDATE**, **DELETE**, and **JOIN** operations.

Table 1: Movies

Movie ID	Movie Name	Language	Genre	Ticket Price (₹)
401	Pushpa 3	Telugu	Action	350
402	Kalki 2898 AD	Telugu	Sci-Fi	400
403	Chhaava	Hindi	Historical	300
404	Housefull 5	Hindi	Comedy	250
405	Coolie	Tamil	Action	320
406	Jailer 2	Tamil	Action	340
407	Kantara Chapter 1	Kannada	Thriller	330
408	Martin	Kannada	Action	290
409	Sitaare Zameen Par	Hindi	Drama	280
410	War 2	Hindi	Action	380
411	Dragon	Tamil	Drama	260
412	HIT 3	Telugu	Crime	310
413	Tourist Family	Tamil	Family	220
414	Retro	Tamil	Romance	240
415	Thug Life	Tamil	Action	360
416	Lucifer 2	Malayalam	Action	330
417	Empuraan	Malayalam	Thriller	350
418	Marco	Malayalam	Crime	300
419	Deva	Hindi	Action	290
420	Bhool Chuk Maaf	Hindi	Comedy	240
421	Kanguva	Tamil	Fantasy	370
422	Vidaamuyarchi	Tamil	Thriller	320
423	Game Changer	Telugu	Political Drama	340
424	Indian 3	Tamil	Action	360
425	Baaghi 4	Hindi	Action	310

Table 2: Bookings

Booking ID	Customer Name	Movie ID	Number of Tickets	Booking Date
501	Aarav Gupta	401	2	2026-03-01
502	Siya Sharma	402	4	2026-03-01
503	Rohan Verma	403	3	2026-03-02
504	Ananya Singh	404	2	2026-03-02
505	Kabir Patel	405	5	2026-03-03
506	Ishita Joshi	406	2	2026-03-03
507	Vihaan Mehta	407	1	2026-03-04
508	Myra Nair	408	3	2026-03-04
509	Arjun Das	409	2	2026-03-05
510	Kiara Roy	410	4	2026-03-05
511	Aditya Kumar	411	2	2026-03-06

512	Aditi Shah	412	3	2026-03-06
513	Dev Yadav	413	4	2026-03-07
514	Riya Jain	414	2	2026-03-07
515	Yash Saini	415	5	2026-03-08
516	Mehak Khan	416	1	2026-03-08
517	Aryan Mishra	417	3	2026-03-09
518	Tanvi Agarwal	418	2	2026-03-09
519	Kunal Malhotra	419	4	2026-03-10
520	Pari Kaur	420	2	2026-03-10
521	Om Vyas	421	5	2026-03-11
522	Diya Rao	422	3	2026-03-11
523	Rudra Saxena	423	2	2026-03-12
524	Saanvi Singh	424	4	2026-03-12
525	Krish Bose	425	1	2026-03-13

Practice Questions

CREATE

- Create a database named **MovieBookingDB**.
- Create the **Movies** table.
- Create the **Bookings** table with an appropriate foreign key.

INSERT

- Insert all 25 movie records.
- Insert all 25 booking records.

UPDATE

- Change the ticket price of **War 2** from ₹380 to ₹400.
- Update the genre of **Housefull 5** from **Comedy** to **Comedy Drama**.
- Change the number of tickets booked by **Siya Sharma** from 4 to 5.
- Update the booking date of **Booking ID 516** to **2026-03-10**.
- Change the movie booked by **Krish Bose** from **Baaghi 4** to **Pushpa 3**.

DELETE

- Delete **Booking ID 525**.
- Delete the movie **Indian 3**.
- Delete the booking made by **Vihaan Mehta**.
- Delete the movie **Dragon**.
- Delete the booking made on **2026-03-03** by **Kabir Patel**.

JOIN

- Display the **customer name** along with the **movie name** they booked.
- Display the **customer name**, **movie language**, and **booking date**.
- Display the **customer name**, **movie genre**, and **number of tickets** booked.
- Display all customers who booked **Hindi** movies.
- Display the **customer name**, **movie name**, and **ticket price**.


```
MySQL 8.0 Command Line CLI x + v
mysql> UPDATE customers
-> SET city = 'Gurugram'
-> WHERE customer_id = 115;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

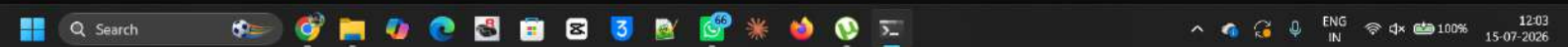
mysql> SELECT *
-> FROM customers
-> WHERE customer_id = 115;
+-----+-----+-----+-----+
| customer_id | customer_name | city      | phone      |
+-----+-----+-----+-----+
| 115         | Mohit Saini   | Gurugram | 9876543215 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> UPDATE customers
-> SET phone = '9296197038'
-> WHERE customer_id = 122;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT *
-> FROM customers
-> WHERE customer_id = 122;
+-----+-----+-----+-----+
| customer_id | customer_name | city      | phone      |
+-----+-----+-----+-----+
| 122         | Divya Rao     | Mysore    | 9296197038 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> UPDATE orders
-> SET quantity = 8
-> WHERE order_id = 1017;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT *
-> FROM customers
-> WHERE customer_id = 135;
```



MySQL 8.0 Command Line CLI

```
mysql> SELECT *
-> FROM customers
-> WHERE customer_id = 135;
Empty set (0.00 sec)

mysql> SELECT *
-> FROM orders
-> WHERE order_id = 1017;
+-----+-----+-----+-----+-----+-----+
| order_id | customer_id | product_name | quantity | price | order_date |
+-----+-----+-----+-----+-----+-----+
| 1017 | 117 | Pendrive | 8 | 900.00 | 2026-01-25 |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> UPDATE orders
-> SET price = 37000
-> WHERE product_name = 'Projector';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE orders
-> SET product_name = 'WiFi Router'
-> WHERE product_name = 'Router';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT * FROM orders
-> WHERE order_id = 1017
-> OR product_name = 'Projector'
-> OR product_name = 'WiFi Router';
+-----+-----+-----+-----+-----+-----+
| order_id | customer_id | product_name | quantity | price | order_date |
+-----+-----+-----+-----+-----+-----+
| 1015 | 115 | WiFi Router | 1 | 2800.00 | 2026-01-23 |
| 1017 | 117 | Pendrive | 8 | 900.00 | 2026-01-25 |
| 1025 | 125 | Projector | 1 | 37000.00 | 2026-02-02 |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

Search

ENG IN 100% 12:03 15-07-2026

```
mysql> DELETE customers
-> FEOM
-> \C
ERROR:
Usage: \C charset_name | charset charset_name
-> \c
mysql> DELETE FROM customers
-> WHERE customer_id = 125;
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key constraint fails (`online_sopping_system`.`orders`, CONSTRAINT `orders_ibfk_1` FOREIGN KEY (`customer_id`) REFERENCES `customers` (`customer_id`))
mysql> DELETE FROM orders
-> WHERE order_id = 1025;
Query OK, 1 row affected (0.01 sec)

mysql> ^C
mysql> DELETE FROM orders
-> WHERE product_name = 'Gaming Mouse';
Query OK, 1 row affected (0.01 sec)

mysql> DELETE FROM customers
-> WHERE customer_name = 'Harsh Vyas';
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key constraint fails (`online_sopping_system`.`orders`, CONSTRAINT `orders_ibfk_1` FOREIGN KEY (`customer_id`) REFERENCES `customers` (`customer_id`))
mysql> DELETE FROM orders
-> WHERE order_date = '2026-01-14';
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM customers;
```

customer_id	customer_name	city	phone
101	Amit Sharma	Delhi	9876543201
102	Priya Verma	Mumbai	9876543202
103	Rahul Singh	Pune	9876543203
104	Sneha Gupta	Jaipur	9876543204
105	Arjun Patel	Ahmedabad	9876543205
106	Neha Joshi	Surat	9876543206
107	Karan Mehta	Chandigarh	9876543207
108	Pooja Nair	Kochi	9876543208
109	Rohan Das	Kolkata	9876543209

```
MySQL 8.0 Command Line CLI x + v
118 | Ritu Agarwal | Agra | 9876543218 |
119 | Akash Malhotra | Ludhiana | 9876543219 |
120 | Simran Kaur | Amritsar | 9876543220 |
121 | Harsh Vyas | Udaipur | 9876543221 |
122 | Divya Rao | Mysore | 9296197038 |
123 | Manish Saxena | Dehradun | 9876543223 |
124 | Kavita Singh | Ranchi | 9876543224 |
125 | Aditya Bose | Bhubaneswar | 9876543225 |
-----+-----
25 rows in set (0.00 sec)

mysql> SELECT * FROM orders;
+-----+-----+-----+-----+-----+-----+
| order_id | customer_id | product_name | quantity | price | order_date |
+-----+-----+-----+-----+-----+-----+
| 1001 | 101 | Laptop | 1 | 65000.00 | 2026-01-10 |
| 1002 | 102 | Smartphone | 2 | 25000.00 | 2026-01-11 |
| 1003 | 103 | Headphones | 3 | 2500.00 | 2026-01-12 |
| 1004 | 104 | Keyboard | 2 | 1800.00 | 2026-01-12 |
| 1005 | 105 | Mouse | 4 | 700.00 | 2026-01-13 |
| 1007 | 107 | Printer | 1 | 12000.00 | 2026-01-15 |
| 1008 | 108 | Tablet | 2 | 28000.00 | 2026-01-16 |
| 1009 | 109 | Webcam | 3 | 2200.00 | 2026-01-17 |
| 1010 | 110 | SSD | 2 | 5000.00 | 2026-01-18 |
| 1011 | 111 | Laptop | 1 | 72000.00 | 2026-01-19 |
| 1012 | 112 | Smartphone | 1 | 30000.00 | 2026-01-20 |
| 1013 | 113 | Smart Watch | 2 | 8500.00 | 2026-01-21 |
| 1014 | 114 | Bluetooth Speaker | 2 | 3500.00 | 2026-01-22 |
| 1015 | 115 | WiFi Router | 1 | 2800.00 | 2026-01-23 |
| 1016 | 116 | Power Bank | 3 | 1800.00 | 2026-01-24 |
| 1017 | 117 | Pendrive | 8 | 900.00 | 2026-01-25 |
| 1018 | 118 | External HDD | 2 | 6500.00 | 2026-01-26 |
| 1020 | 120 | Mechanical Keyboard | 1 | 5500.00 | 2026-01-28 |
| 1021 | 121 | Laptop Bag | 2 | 1800.00 | 2026-01-29 |
| 1022 | 122 | UPS | 1 | 6500.00 | 2026-01-30 |
| 1023 | 123 | Graphics Card | 1 | 42000.00 | 2026-01-31 |
| 1024 | 124 | CPU Cabinet | 1 | 4800.00 | 2026-02-01 |
+-----+-----+-----+-----+-----+-----+
22 rows in set (0.00 sec)

mysql>
```

```
MySQL 8.0 Command Line CLI x + v
mysql> CREATE database Hospital_Management_System;
Query OK, 1 row affected (0.01 sec)

mysql> CREATE TABLE doctors (
->     doctor_id INT PRIMARY KEY,
->     doctor_name VARCHAR(100) NOT NULL,
->     specialization VARCHAR(50) NOT NULL,
->     experience_years INT NOT NULL
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO doctors (doctor_id, doctor_name, specialization, experience_years) VALUES
-> (201, 'Dr. Amit Sharma', 'Cardiologist', 15),
-> (202, 'Dr. Priya Verma', 'Neurologist', 12),
-> (203, 'Dr. Rahul Singh', 'Orthopedic', 10),
-> (204, 'Dr. Sneha Gupta', 'Pediatrician', 8),
-> (205, 'Dr. Arjun Patel', 'Dermatologist', 9),
-> (206, 'Dr. Neha Joshi', 'Gynecologist', 14),
-> (207, 'Dr. Karan Mehta', 'ENT Specialist', 11),
-> (208, 'Dr. Pooja Mair', 'Ophthalmologist', 7),
-> (209, 'Dr. Rohan Das', 'Psychiatrist', 13),
-> (210, 'Dr. Anjali Roy', 'General Physician', 16),
-> (211, 'Dr. Vivek Kumar', 'Urologist', 9),
-> (212, 'Dr. Meera Shah', 'Oncologist', 17),
-> (213, 'Dr. Deepak Yadav', 'Nephrologist', 12),
-> (214, 'Dr. Nisha Jain', 'Pulmonologist', 10),
-> (215, 'Dr. Mohit Saini', 'Endocrinologist', 8),
-> (216, 'Dr. Aisha Khan', 'Gastroenterologist', 11),
-> (217, 'Dr. Saurabh Mishra', 'Rheumatologist', 14),
-> (218, 'Dr. Ritu Agarwal', 'Surgeon', 18),
-> (219, 'Dr. Akash Malhotra', 'Cardiologist', 6),
-> (220, 'Dr. Simran Kaur', 'Pediatrician', 5),
-> (221, 'Dr. Harsh Vyas', 'Dentist', 9),
-> (222, 'Dr. Divya Rao', 'Neurologist', 13),
-> (223, 'Dr. Manish Saxena', 'Orthopedic', 7),
-> (224, 'Dr. Kavita Singh', 'General Physician', 15),
-> (225, 'Dr. Aditya Bose', 'Dermatologist', 8);
Query OK, 25 rows affected (0.01 sec)
Records: 25 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM doctors;
```

mysql> SELECT * FROM doctors;

doctor_id	doctor_name	specialization	experience_years
201	Dr. Amit Sharma	Cardiologist	15
202	Dr. Priya Verma	Neurologist	12
203	Dr. Rahul Singh	Orthopedic	10
204	Dr. Sneha Gupta	Pediatrician	8
205	Dr. Arjun Patel	Dermatologist	9
206	Dr. Neha Joshi	Gynecologist	14
207	Dr. Karan Mehta	ENT Specialist	11
208	Dr. Pooja Nair	Ophthalmologist	7
209	Dr. Rohan Das	Psychiatrist	13
210	Dr. Anjali Roy	General Physician	16
211	Dr. Vivek Kumar	Urologist	9
212	Dr. Meera Shah	Oncologist	17
213	Dr. Deepak Yadav	Nephrologist	12
214	Dr. Nisha Jain	Pulmonologist	10
215	Dr. Mohit Saini	Endocrinologist	8
216	Dr. Aisha Khan	Gastroenterologist	11
217	Dr. Saurabh Mishra	Rheumatologist	14
218	Dr. Ritu Agarwal	Surgeon	18
219	Dr. Akash Malhotra	Cardiologist	6
220	Dr. Simran Kaur	Pediatrician	5
221	Dr. Harsh Vyas	Dentist	9
222	Dr. Divya Rao	Neurologist	13
223	Dr. Manish Saxena	Orthopedic	7
224	Dr. Kavita Singh	General Physician	15
225	Dr. Aditya Bose	Dermatologist	8

25 rows in set (0.00 sec)

```
mysql> CREATE TABLE patients (  
-> patient_id INT PRIMARY KEY,  
-> patient_name VARCHAR(100) NOT NULL,  
-> doctor_id INT NOT NULL,  
-> disease VARCHAR(100) NOT NULL,  
-> admission_date DATE NOT NULL,  
-> FOREIGN KEY (doctor_id) REFERENCES doctors(doctor_id)  
-> );
```



```
mysql> INSERT INTO patients (patient_id, patient_name, doctor_id, disease, admission_date) VALUES
-> (301, 'Aarav Gupta', 201, 'Heart Disease', '2026-02-01'),
-> (302, 'Siya Sharma', 202, 'Migraine', '2026-02-02'),
-> (303, 'Rohan Verma', 203, 'Fracture', '2026-02-03'),
-> (304, 'Ananya Singh', 204, 'Viral Fever', '2026-02-04'),
-> (305, 'Kabir Patel', 205, 'Skin Allergy', '2026-02-05'),
-> (306, 'Ishita Joshi', 206, 'Pregnancy Checkup', '2026-02-06'),
-> (307, 'Vihaan Mehta', 207, 'Ear Infection', '2026-02-07'),
-> (308, 'Myra Nair', 208, 'Eye Infection', '2026-02-08'),
-> (309, 'Arjun Das', 209, 'Anxiety', '2026-02-09'),
-> (310, 'Kiara Roy', 210, 'Diabetes', '2026-02-10'),
-> (311, 'Aditya Kumar', 211, 'Kidney Stone', '2026-02-11'),
-> (312, 'Aditi Shah', 212, 'Cancer Screening', '2026-02-12'),
-> (313, 'Dev Yadav', 213, 'Kidney Infection', '2026-02-13'),
-> (314, 'Riya Jain', 214, 'Asthma', '2026-02-14'),
-> (315, 'Yash Saini', 215, 'Thyroid', '2026-02-15'),
-> (316, 'Mehak Khan', 216, 'Ulcer', '2026-02-16'),
-> (317, 'Aryan Mishra', 217, 'Arthritis', '2026-02-17'),
-> (318, 'Tanvi Agarwal', 218, 'Appendicitis', '2026-02-18'),
-> (319, 'Kunal Malhotra', 219, 'Heart Checkup', '2026-02-19'),
-> (320, 'Pari Kaur', 220, 'Cold & Cough', '2026-02-20'),
-> (321, 'Om Vyas', 221, 'Tooth Decay', '2026-02-21'),
-> (322, 'Diya Rao', 222, 'Epilepsy', '2026-02-22'),
-> (323, 'Rudra Saxena', 223, 'Knee Pain', '2026-02-23'),
-> (324, 'Saanvi Singh', 224, 'High Fever', '2026-02-24'),
-> (325, 'Krish Bose', 225, 'Acne', '2026-02-25');
Query OK, 25 rows affected (0.01 sec)
Records: 25 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM patients
->
-> \c
mysql> SELECT * FROM patients;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'F
ORM patients' at line 1
mysql> SELECT * FROM patients;
+-----+-----+-----+-----+-----+
| patient_id | patient_name | doctor_id | disease | admission_date |
+-----+-----+-----+-----+-----+

```

```
MySQL 8.0 Command Line CLI x + v
mysql> SELECT * FROM patients;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'FROM patients' at line 1
mysql> SELECT * FROM patients;
```

patient_id	patient_name	doctor_id	disease	admission_date
301	Aarav Gupta	201	Heart Disease	2026-02-01
302	Siya Sharma	202	Migraine	2026-02-02
303	Rohan Verma	203	Fracture	2026-02-03
304	Ananya Singh	204	Viral Fever	2026-02-04
305	Kabir Patel	205	Skin Allergy	2026-02-05
306	Ishita Joshi	206	Pregnancy Checkup	2026-02-06
307	Vihaan Mehta	207	Ear Infection	2026-02-07
308	Myra Mair	208	Eye Infection	2026-02-08
309	Arjun Das	209	Anxiety	2026-02-09
310	Kiara Roy	210	Diabetes	2026-02-10
311	Aditya Kumar	211	Kidney Stone	2026-02-11
312	Aditi Shah	212	Cancer Screening	2026-02-12
313	Dev Yadav	213	Kidney Infection	2026-02-13
314	Riya Jain	214	Asthma	2026-02-14
315	Yash Saini	215	Thyroid	2026-02-15
316	Mehak Khan	216	Ulcer	2026-02-16
317	Aryan Mishra	217	Arthritis	2026-02-17
318	Tanvi Agarwal	218	Appendicitis	2026-02-18
319	Kunal Malhotra	219	Heart Checkup	2026-02-19
320	Pari Kaur	220	Cold & Cough	2026-02-20
321	Om Vyas	221	Tooth Decay	2026-02-21
322	Diya Rao	222	Epilepsy	2026-02-22
323	Rudra Saxena	223	Knee Pain	2026-02-23
324	Saanvi Singh	224	High Fever	2026-02-24
325	Krish Bose	225	Acne	2026-02-25

```
25 rows in set (0.00 sec)

mysql> UPDATE doctors
-> SET specialization = 'Cosmetologist'
-> WHERE doctor_id = 225;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

Windows Search

ENG IN 100% 12:23 15-07-2026


```
MySQL 8.0 Command Line CLI x + v

mysql> UPDATE doctors
-> SET experience_years = 7
-> WHERE doctor_id = 220;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE patients
-> SET disease = 'Depression'
-> WHERE patient_id = 309;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE patients
-> SET admission_date = '2026-02-22'
-> WHERE patient_id = 320;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE patients
-> SET doctor_id = 203
-> WHERE patient_id = 323;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql>
mysql> SELECT * FROM doctors
-> WHERE doctor_id IN (220, 225);
+-----+-----+-----+-----+
| doctor_id | doctor_name | specialization | experience_years |
+-----+-----+-----+-----+
| 220 | Dr. Simran Kaur | Pediatrician | 7 |
| 225 | Dr. Aditya Bose | Cosmetologist | 8 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM patients
-> WHERE patient_id IN (309, 320, 323);
+-----+-----+-----+-----+-----+
| patient_id | patient_name | doctor_id | disease | admission_date |
+-----+-----+-----+-----+-----+
```

```
MySQL 8.0 Command Line CLI x + v

mysql> DELETE FROM patients
-> WHERE patient_id = 325;
Query OK, 1 row affected (0.01 sec)

mysql> DELETE FROM doctors
-> WHERE doctor_id = 225;
Query OK, 1 row affected (0.03 sec)

mysql> DELETE FROM patients
-> WHERE patient_name = 'Pari Kaur';
Query OK, 1 row affected (0.01 sec)

mysql> DELETE FROM doctors
-> WHERE doctor_name = 'Dr. Harsh Vyas';
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key constraint fails ('online_sopping_system`.`patients`, CONSTRAINT `patients_ibfk_1` FOREIGN KEY (`doctor_id`) REFERENCES `doctors` (`doctor_id`))
mysql> DELETE FROM patients
-> WHERE doctor_id = 221;
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> DELETE FROM doctors
-> WHERE doctor_name = 'Dr. Harsh Vyas';
Query OK, 1 row affected (0.01 sec)

mysql> DELETE FROM patients
-> WHERE admission_date = '2026-02-06';
Query OK, 1 row affected (0.01 sec)

mysql> DELETE FROM patients
-> WHERE doctor_id = 221;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> DELETE FROM doctors
-> WHERE doctor_name = 'Dr. Harsh Vyas';
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> DELETE FROM patients
-> WHERE doctor_id = 221;
```

MySQL 8.0 Command Line CLI

Query OK, 0 rows affected (0.00 sec)

mysql> DELETE FROM patients
-> WHERE doctor_id = 221;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> DELETE FROM doctors
-> WHERE doctor_name = 'Dr. Harsh Vyas';
Query OK, 0 rows affected (0.00 sec)

mysql> DELETE FROM patients
-> WHERE admission_date = '2026-02-06';
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM doctors;

doctor_id	doctor_name	specialization	experience_years
201	Dr. Amit Sharma	Cardiologist	15
202	Dr. Priya Verma	Neurologist	12
203	Dr. Rahul Singh	Orthopedic	10
204	Dr. Sneha Gupta	Pediatrician	8
205	Dr. Arjun Patel	Dermatologist	9
206	Dr. Neha Joshi	Gynecologist	14
207	Dr. Karan Mehta	ENT Specialist	11
208	Dr. Pooja Nair	Ophthalmologist	7
209	Dr. Rohan Das	Psychiatrist	13
210	Dr. Anjali Roy	General Physician	16
211	Dr. Vivek Kumar	Urologist	9
212	Dr. Meera Shah	Oncologist	17
213	Dr. Deepak Yadav	Nephrologist	12
214	Dr. Nisha Jain	Pulmonologist	10
215	Dr. Mohit Saini	Endocrinologist	8
216	Dr. Aisha Khan	Gastroenterologist	11
217	Dr. Saurabh Mishra	Rheumatologist	14
218	Dr. Ritu Agarwal	Surgeon	18
219	Dr. Akash Malhotra	Cardiologist	6
220	Dr. Simran Kaur	Pediatrician	7
222	Dr. Divya Rao	Neurologist	13
223	Dr. Manish Saxena	Orthopedic	7

Q Search

ENG IN 100% 12:28 15-07-2026

MySQL 8.0 Command Line CLI

216	Dr. Aisha Khan	Gastroenterologist	11
217	Dr. Saurabh Mishra	Rheumatologist	14
218	Dr. Ritu Agarwal	Surgeon	18
219	Dr. Akash Malhotra	Cardiologist	6
220	Dr. Simran Kaur	Pediatrician	7
222	Dr. Divya Rao	Neurologist	13
223	Dr. Manish Saxena	Orthopedic	7
224	Dr. Kavita Singh	General Physician	15

23 rows in set (0.00 sec)

```
mysql>
mysql> SELECT * FROM patients;
```

patient_id	patient_name	doctor_id	disease	admission_date
301	Aarav Gupta	201	Heart Disease	2026-02-01
302	Siya Sharma	202	Migraine	2026-02-02
303	Rohan Verma	203	Fracture	2026-02-03
304	Ananya Singh	204	Viral Fever	2026-02-04
305	Kabir Patel	205	Skin Allergy	2026-02-05
307	Vihaan Mehta	207	Ear Infection	2026-02-07
308	Myra Nair	208	Eye Infection	2026-02-08
309	Arjun Das	209	Depression	2026-02-09
310	Kiara Roy	210	Diabetes	2026-02-10
311	Aditya Kumar	211	Kidney Stone	2026-02-11
312	Aditi Shah	212	Cancer Screening	2026-02-12
313	Dev Yadav	213	Kidney Infection	2026-02-13
314	Riya Jain	214	Asthma	2026-02-14
315	Yash Saini	215	Thyroid	2026-02-15
316	Mehak Khan	216	Ulcer	2026-02-16
317	Aryan Mishra	217	Arthritis	2026-02-17
318	Tanvi Agarwal	218	Appendicitis	2026-02-18
319	Kunal Malhotra	219	Heart Checkup	2026-02-19
322	Diya Rao	222	Epilepsy	2026-02-22
323	Rudra Saxena	203	Knee Pain	2026-02-23
324	Saanvi Singh	224	High Fever	2026-02-24

21 rows in set (0.00 sec)

```
mysql>
```

Search

ENG IN 100% 12:29 15-07-2026

MySQL 8.0 Command Line CLI

```
mysql> SELECT p.patient_name, d.doctor_name
-> FROM patients p
-> INNER JOIN doctors d
-> ON p.doctor_id = d.doctor_id;
```

patient_name	doctor_name
Aarav Gupta	Dr. Amit Sharma
Siya Sharma	Dr. Priya Verma
Rohan Verma	Dr. Rahul Singh
Ananya Singh	Dr. Sneha Gupta
Kabir Patel	Dr. Arjun Patel
Vihaan Mehta	Dr. Karan Mehta
Myra Nair	Dr. Pooja Nair
Arjun Das	Dr. Rohan Das
Kiara Roy	Dr. Anjali Roy
Aditya Kumar	Dr. Vivek Kumar
Aditi Shah	Dr. Meera Shah
Dev Yadav	Dr. Deepak Yadav
Riya Jain	Dr. Nisha Jain
Yash Saini	Dr. Mohit Saini
Mehak Khan	Dr. Aisha Khan
Aryan Mishra	Dr. Saurabh Mishra
Tanvi Agarwal	Dr. Ritu Agarwal
Kunal Malhotra	Dr. Akash Malhotra
Diya Rao	Dr. Divya Rao
Rudra Saxena	Dr. Rahul Singh
Saanvi Singh	Dr. Kavita Singh

21 rows in set (0.00 sec)

```
mysql> SELECT p.patient_name,
-> p.disease,
-> d.specialization
-> FROM patients p
-> INNER JOIN doctors d
-> ON p.doctor_id = d.doctor_id;
```

patient_name	disease	specialization
--------------	---------	----------------

Search

ENG IN 100% 12:31 15-07-2026

MySQL 8.0 Command Line CLI

patient_name	disease	specialization
Aarav Gupta	Heart Disease	Cardiologist
Siya Sharma	Migraine	Neurologist
Rohan Verma	Fracture	Orthopedic
Ananya Singh	Viral Fever	Pediatrician
Kabir Patel	Skin Allergy	Dermatologist
Vihaan Mehta	Ear Infection	ENT Specialist
Myra Nair	Eye Infection	Ophthalmologist
Arjun Das	Depression	Psychiatrist
Kiara Roy	Diabetes	General Physician
Aditya Kumar	Kidney Stone	Urologist
Aditi Shah	Cancer Screening	Oncologist
Dev Yadav	Kidney Infection	Nephrologist
Riya Jain	Asthma	Pulmonologist
Yash Saini	Thyroid	Endocrinologist
Mehak Khan	Ulcer	Gastroenterologist
Aryan Mishra	Arthritis	Rheumatologist
Tanvi Agarwal	Appendicitis	Surgeon
Kunal Malhotra	Heart Checkup	Cardiologist
Diya Rao	Epilepsy	Neurologist
Rudra Saxena	Knee Pain	Orthopedic
Saanvi Singh	High Fever	General Physician

21 rows in set (0.00 sec)

```
mysql> SELECT p.patient_name,
->         d.doctor_name,
->         p.admission_date
-> FROM patients p
-> INNER JOIN doctors d
-> ON p.doctor_id = d.doctor_id;
```

patient_name	doctor_name	admission_date
Aarav Gupta	Dr. Amit Sharma	2026-02-01
Siya Sharma	Dr. Priya Verma	2026-02-02
Rohan Verma	Dr. Rahul Singh	2026-02-03
Ananya Singh	Dr. Sneha Gupta	2026-02-04
Kabir Patel	Dr. Arjun Patel	2026-02-05

Search

12:32
15-07-2026

MySQL 8.0 Command Line CLI

21 rows in set (0.00 sec)

mysql> SELECT p.patient_name,
-> d.doctor_name,
-> p.admission_date
-> FROM patients p
-> INNER JOIN doctors d
-> ON p.doctor_id = d.doctor_id;

patient_name	doctor_name	admission_date
Aarav Gupta	Dr. Amit Sharma	2026-02-01
Siya Sharma	Dr. Priya Verma	2026-02-02
Rohan Verma	Dr. Rahul Singh	2026-02-03
Ananya Singh	Dr. Sneha Gupta	2026-02-04
Kabir Patel	Dr. Arjun Patel	2026-02-05
Vihaan Mehta	Dr. Karan Mehta	2026-02-07
Myra Nair	Dr. Pooja Nair	2026-02-08
Arjun Das	Dr. Rohan Das	2026-02-09
Kiara Roy	Dr. Anjali Roy	2026-02-10
Aditya Kumar	Dr. Vivek Kumar	2026-02-11
Aditi Shah	Dr. Meera Shah	2026-02-12
Dev Yadav	Dr. Deepak Yadav	2026-02-13
Riya Jain	Dr. Nisha Jain	2026-02-14
Yash Saini	Dr. Mohit Saini	2026-02-15
Mehak Khan	Dr. Aisha Khan	2026-02-16
Aryan Mishra	Dr. Saurabh Mishra	2026-02-17
Tanvi Agarwal	Dr. Ritu Agarwal	2026-02-18
Kunal Malhotra	Dr. Akash Malhotra	2026-02-19
Diya Rao	Dr. Divya Rao	2026-02-22
Rudra Saxena	Dr. Rahul Singh	2026-02-23
Saanvi Singh	Dr. Kavita Singh	2026-02-24

21 rows in set (0.00 sec)

mysql> SELECT d.doctor_name,
-> d.experience_years,
-> p.patient_name
-> FROM doctors d
-> INNER JOIN patients p

Search

ENG IN 100% 12:32 15-07-2026

MySQL 8.0 Command Line CLI

doctor_name	experience_years	patient_name
Dr. Amit Sharma	15	Aarav Gupta
Dr. Priya Verma	12	Siya Sharma
Dr. Rahul Singh	10	Rohan Verma
Dr. Sneha Gupta	8	Ananya Singh
Dr. Arjun Patel	9	Kabir Patel
Dr. Karan Mehta	11	Vihaan Mehta
Dr. Pooja Nair	7	Myra Nair
Dr. Rohan Das	13	Arjun Das
Dr. Anjali Roy	16	Kiara Roy
Dr. Vivek Kumar	9	Aditya Kumar
Dr. Meera Shah	17	Aditi Shah
Dr. Deepak Yadav	12	Dev Yadav
Dr. Nisha Jain	10	Riya Jain
Dr. Mohit Saini	8	Yash Saini
Dr. Aisha Khan	11	Mehak Khan
Dr. Saurabh Mishra	14	Aryan Mishra
Dr. Ritu Agarwal	18	Tanvi Agarwal
Dr. Akash Malhotra	6	Kunal Malhotra
Dr. Divya Rao	13	Diya Rao
Dr. Rahul Singh	10	Rudra Saxena
Dr. Kavita Singh	15	Saanvi Singh

21 rows in set (0.00 sec)

```
mysql> SELECT d.doctor_name,
->         d.experience_years,
->         p.patient_name
-> FROM doctors d
-> INNER JOIN patients p
-> ON d.doctor_id = p.doctor_id;
```

doctor_name	experience_years	patient_name
Dr. Amit Sharma	15	Aarav Gupta
Dr. Priya Verma	12	Siya Sharma
Dr. Rahul Singh	10	Rohan Verma
Dr. Sneha Gupta	8	Ananya Singh
Dr. Arjun Patel	9	Kabir Patel

Search

12:32
15-07-2026

MySQL 8.0 Command Line CLI

Dr. Amit Sharma	15	Aarav Gupta
Dr. Priya Verma	12	Siya Sharma
Dr. Rahul Singh	10	Rohan Verma
Dr. Sneha Gupta	8	Ananya Singh
Dr. Arjun Patel	9	Kabir Patel
Dr. Karan Mehta	11	Vihaan Mehta
Dr. Pooja Nair	7	Myra Nair
Dr. Rohan Das	13	Arjun Das
Dr. Anjali Roy	16	Kiara Roy
Dr. Vivek Kumar	9	Aditya Kumar
Dr. Meera Shah	17	Aditi Shah
Dr. Deepak Yadav	12	Dev Yadav
Dr. Nisha Jain	10	Riya Jain
Dr. Mohit Saini	8	Yash Saini
Dr. Aisha Khan	11	Mehak Khan
Dr. Saurabh Mishra	14	Aryan Mishra
Dr. Ritu Agarwal	18	Tanvi Agarwal
Dr. Akash Malhotra	6	Kunal Malhotra
Dr. Divya Rao	13	Diya Rao
Dr. Rahul Singh	10	Rudra Saxena
Dr. Kavita Singh	15	Saanvi Singh

21 rows in set (0.00 sec)

```
mysql> SELECT p.patient_name,
->      p.disease,
->      d.doctor_name
-> FROM patients p
-> INNER JOIN doctors d
-> ON p.doctor_id = d.doctor_id
-> WHERE d.specialization = 'Cardiologist';
```

patient_name	disease	doctor_name
Aarav Gupta	Heart Disease	Dr. Amit Sharma
Kunal Malhotra	Heart Checkup	Dr. Akash Malhotra

2 rows in set (0.00 sec)

```
mysql>
```

Search

ENG IN 100% 12:32 15-07-2026

```
MySQL 8.0 Command Line Cli  x  +  v

mysql> CREATE TABLE departments (
->     department_id INT PRIMARY KEY,
->     department_name VARCHAR(100) NOT NULL,
->     hod_name VARCHAR(100) NOT NULL,
->     building VARCHAR(20) NOT NULL
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO departments (department_id, department_name, hod_name, building) VALUES
-> (301, 'Computer Science', 'Dr. Amit Sharma', 'Block A'),
-> (302, 'Information Technology', 'Dr. Priya Verma', 'Block B'),
-> (303, 'Electronics', 'Dr. Rahul Singh', 'Block C'),
-> (304, 'Mechanical', 'Dr. Sneha Gupta', 'Block D'),
-> (305, 'Civil', 'Dr. Arjun Patel', 'Block E'),
-> (306, 'Electrical', 'Dr. Neha Joshi', 'Block F'),
-> (307, 'Artificial Intelligence', 'Dr. Karan Mehta', 'Block G'),
-> (308, 'Data Science', 'Dr. Pooja Nair', 'Block H'),
-> (309, 'Biotechnology', 'Dr. Rohan Das', 'Block I'),
-> (310, 'Chemical', 'Dr. Anjali Roy', 'Block J'),
-> (311, 'Commerce', 'Dr. Vivek Kumar', 'Block K'),
-> (312, 'Management', 'Dr. Meera Shah', 'Block L'),
-> (313, 'Mathematics', 'Dr. Deepak Yadav', 'Block M'),
-> (314, 'Physics', 'Dr. Nisha Jain', 'Block N'),
-> (315, 'Chemistry', 'Dr. Mohit Saini', 'Block O'),
-> (316, 'English', 'Dr. Aisha Khan', 'Block P'),
-> (317, 'Economics', 'Dr. Saurabh Mishra', 'Block Q'),
-> (318, 'Psychology', 'Dr. Ritu Agarwal', 'Block R'),
-> (319, 'History', 'Dr. Akash Malhotra', 'Block S'),
-> (320, 'Geography', 'Dr. Simran Kaur', 'Block T'),
-> (321, 'Sociology', 'Dr. Harsh Vyas', 'Block U'),
-> (322, 'Political Science', 'Dr. Divya Rao', 'Block V'),
-> (323, 'Finance', 'Dr. Manish Saxena', 'Block W'),
-> (324, 'Marketing', 'Dr. Kavita Singh', 'Block X'),
-> (325, 'Human Resources', 'Dr. Aditya Bose', 'Block Y');
Query OK, 25 rows affected (0.01 sec)
Records: 25  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE students (
->     student_id INT PRIMARY KEY,
->     student_name VARCHAR(100) NOT NULL,
```


MySQL 8.0 Command Line CLI

```
mysql> SELECT * FROM departments;
```

department_id	department_name	hod_name	building
301	Computer Science	Dr. Amit Sharma	Block A
302	Information Technology	Dr. Priya Verma	Block B
303	Electronics	Dr. Rahul Singh	Block C
304	Mechanical	Dr. Sneha Gupta	Block D
305	Civil	Dr. Arjun Patel	Block E
306	Electrical	Dr. Neha Joshi	Block F
307	Artificial Intelligence	Dr. Karan Mehta	Block G
308	Data Science	Dr. Pooja Nair	Block H
309	Biotechnology	Dr. Rohan Das	Block I
310	Chemical	Dr. Anjali Roy	Block J
311	Commerce	Dr. Vivek Kumar	Block K
312	Management	Dr. Meera Shah	Block L
313	Mathematics	Dr. Deepak Yadav	Block M
314	Physics	Dr. Nisha Jain	Block N
315	Chemistry	Dr. Mohit Saini	Block O
316	English	Dr. Aisha Khan	Block P
317	Economics	Dr. Saurabh Mishra	Block Q
318	Psychology	Dr. Ritu Agarwal	Block R
319	History	Dr. Akash Malhotra	Block S
320	Geography	Dr. Simran Kaur	Block T
321	Sociology	Dr. Harsh Vyas	Block U
322	Political Science	Dr. Divya Rao	Block V
323	Finance	Dr. Manish Saxena	Block W
324	Marketing	Dr. Kavita Singh	Block X
325	Human Resources	Dr. Aditya Bose	Block Y

25 rows in set (0.00 sec)

```
mysql> SELECT * FROM students;
```

student_id	student_name	department_id	semester	cgpa
401	Aarav Gupta	301	2	8.5
402	Siya Sharma	302	4	8.2
403	Rohan Verma	303	6	7.8
404	Ananya Singh	304	3	8.9
405	Kabir Patel	305	5	7.4

Search

ENG IN 100% 12:40 15-07-2026

```
MySQL 8.0 Command Line CLI x + v
```

321	Sociology	Dr. Harsh Vyas	Block U
322	Political Science	Dr. Divya Rao	Block V
323	Finance	Dr. Manish Saxena	Block W
324	Marketing	Dr. Kavita Singh	Block X
325	Human Resources	Dr. Aditya Bose	Block Y

25 rows in set (0.00 sec)

```
mysql> SELECT * FROM students;
```

student_id	student_name	department_id	semester	cgpa
401	Aarav Gupta	301	2	8.5
402	Siya Sharma	302	4	8.2
403	Rohan Verma	303	6	7.8
404	Ananya Singh	304	3	8.9
405	Kabir Patel	305	5	7.4
406	Ishita Joshi	306	1	9.1
407	Vihaan Mehta	307	7	8.7
408	Myra Nair	308	2	8.3
409	Arjun Das	309	4	7.6
410	Kiara Roy	310	8	8.8
411	Aditya Kumar	311	5	7.9
412	Aditi Shah	312	6	9.0
413	Dev Yadav	313	3	8.1
414	Riya Jain	314	2	7.5
415	Yash Saini	315	4	8.6
416	Mehak Khan	316	1	9.2
417	Aryan Mishra	317	7	8.0
418	Tanvi Agarwal	318	5	8.4
419	Kunal Malhotra	319	6	7.7
420	Pari Kaur	320	3	8.5
421	Om Vyas	321	2	7.9
422	Diya Rao	322	8	8.8
423	Rudra Saxena	323	4	8.1
424	Saanvi Singh	324	5	7.8
425	Krish Bose	325	6	9.3

25 rows in set (0.00 sec)

```
mysql>
```

Windows Taskbar

Search, File Explorer, Edge, and other icons

System tray: ENG IN, 100%, 12:40, 15-07-2026

```
mysql> UPDATE departments
  -> SET hod_name = 'Dr. Rakesh Gupta'
  -> WHERE department_name = 'Artificial Intelligence';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE departments
  -> SET building = 'Block Z'
  -> WHERE department_name = 'Marketing';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE students
  -> SET cgpa = 8.6
  -> WHERE student_id = 417;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE students
  -> SET semester = 2
  -> WHERE student_id = 406;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE students
  -> SET department_id = 323
  -> WHERE student_id = 424;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM departments
  -> WHERE department_id IN (307, 324);
+-----+-----+-----+-----+
| department_id | department_name | hod_name | building |
+-----+-----+-----+-----+
|          307 | Artificial Intelligence | Dr. Rakesh Gupta | Block G |
|          324 | Marketing | Dr. Kavita Singh | Block Z |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
MySQL 8.0 Command Line CLI x + v
Query OK, 1 row affected (0.02 sec)

mysql> DELETE FROM departments
-> WHERE department_name = 'Human Resources';
Query OK, 1 row affected (0.01 sec)

mysql> DELETE FROM students
-> WHERE student_name = 'Pari Kaur';
Query OK, 1 row affected (0.01 sec)

mysql> DELETE FROM departments
-> WHERE department_name = 'History';
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key constraint fails (`online_sopping_system`.`students`, CONSTRAINT `students_ibfk_1` FOREIGN KEY (`department_id`) REFERENCES `departments` (`department_id`))
mysql> DELETE FROM students
-> WHERE department_id = 319;
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> DELETE FROM departments
-> WHERE department_name = 'History';
Query OK, 1 row affected (0.01 sec)

mysql> DELETE FROM students
-> WHERE student_id = 411;
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM students;
```

student_id	student_name	department_id	semester	cgpa
401	Aarav Gupta	301	2	8.5
402	Siya Sharma	302	4	8.2
403	Rohan Verma	303	6	7.8
404	Ananya Singh	304	3	8.9
405	Kabir Patel	305	5	7.4
406	Ishita Joshi	306	2	9.1
407	Vihaan Mehta	307	7	8.7
408	Myra Nair	308	2	8.3
409	Arjun Das	309	4	7.6
410	Kiara Roy	310	8	8.8

Windows taskbar: Search, File Explorer, Edge, VS Code, WhatsApp, Telegram, 3, Mail, Calendar, 100% battery, 12:46 15-07-2026

MySQL 8.0 Command Line CLI

Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM students;

student_id	student_name	department_id	semester	cgpa
401	Aarav Gupta	301	2	8.5
402	Siya Sharma	302	4	8.2
403	Rohan Verma	303	6	7.8
404	Ananya Singh	304	3	8.9
405	Kabir Patel	305	5	7.4
406	Ishita Joshi	306	2	9.1
407	Vihaan Mehta	307	7	8.7
408	Myra Nair	308	2	8.3
409	Anjun Das	309	4	7.6
410	Kiara Roy	310	8	8.8
412	Aditi Shah	312	6	9.0
413	Dev Yadav	313	3	8.1
414	Riya Jain	314	2	7.5
415	Yash Saini	315	4	8.6
416	Mehak Khan	316	1	9.2
417	Aryan Mishra	317	7	8.6
418	Tanvi Agarwal	318	5	8.4
421	Om Vyas	321	2	7.9
422	Diya Rao	322	8	8.8
423	Rudra Saxena	323	4	8.1
424	Saanvi Singh	323	5	7.8

21 rows in set (0.00 sec)

mysql>

mysql> SELECT * FROM departments;

department_id	department_name	hod_name	building
301	Computer Science	Dr. Amit Sharma	Block A
302	Information Technology	Dr. Priya Verma	Block B
303	Electronics	Dr. Rahul Singh	Block C
304	Mechanical	Dr. Sneha Gupta	Block D
305	Civil	Dr. Arjun Patel	Block E
306	Electrical	Dr. Neha Joshi	Block F

Search

12:47

15-07-2026

MySQL 8.0 Command Line CLI

417	Aryan Mishra	317	7	8.6
418	Tanvi Agarwal	318	5	8.4
421	Om Vyas	321	2	7.9
422	Diya Rao	322	8	8.8
423	Rudra Saxena	323	4	8.1
424	Saanvi Singh	323	5	7.8

21 rows in set (0.00 sec)

```
mysql>
mysql> SELECT * FROM departments;
```

department_id	department_name	hod_name	building
301	Computer Science	Dr. Amit Sharma	Block A
302	Information Technology	Dr. Priya Verma	Block B
303	Electronics	Dr. Rahul Singh	Block C
304	Mechanical	Dr. Sneha Gupta	Block D
305	Civil	Dr. Arjun Patel	Block E
306	Electrical	Dr. Meha Joshi	Block F
307	Artificial Intelligence	Dr. Rakesh Gupta	Block G
308	Data Science	Dr. Pooja Nair	Block H
309	Biotechnology	Dr. Rohan Das	Block I
310	Chemical	Dr. Anjali Roy	Block J
311	Commerce	Dr. Vivek Kumar	Block K
312	Management	Dr. Meera Shah	Block L
313	Mathematics	Dr. Deepak Yadav	Block M
314	Physics	Dr. Nisha Jain	Block N
315	Chemistry	Dr. Mohit Saini	Block O
316	English	Dr. Aisha Khan	Block P
317	Economics	Dr. Saurabh Mishra	Block Q
318	Psychology	Dr. Ritu Agarwal	Block R
320	Geography	Dr. Simran Kaur	Block T
321	Sociology	Dr. Harsh Vyas	Block U
322	Political Science	Dr. Divya Rao	Block V
323	Finance	Dr. Manish Saxena	Block W
324	Marketing	Dr. Kavita Singh	Block Z

23 rows in set (0.00 sec)

```
mysql> -|
```

Search

15-07-2026

12:47

ENG IN

100%

```
MySQL 8.0 Command Line CLI x + -
+-----+-----+-----+-----+
21 rows in set (0.00 sec)

mysql> CREATE DATABASE movieBooлингDB;
Query OK, 1 row affected (0.01 sec)

mysql> CREATE TABLE movies (
->   movie_id INT PRIMARY KEY,
->   movie_name VARCHAR(100) NOT NULL,
->   language VARCHAR(30) NOT NULL,
->   genre VARCHAR(50) NOT NULL,
->   ticket_price DECIMAL(8,2) NOT NULL
-> );
Query OK, 0 rows affected (0.10 sec)

mysql> INSERT INTO movies (movie_id, movie_name, language, genre, ticket_price) VALUES
-> (401, 'Pushpa 3', 'Telugu', 'Action', 350),
-> (402, 'Kalki 2898 AD', 'Telugu', 'Sci-Fi', 400),
-> (403, 'Chhaava', 'Hindi', 'Historical', 300),
-> (404, 'Housefull 5', 'Hindi', 'Comedy', 250),
-> (405, 'Coolie', 'Tamil', 'Action', 320),
-> (406, 'Jailer 2', 'Tamil', 'Action', 340),
-> (407, 'Kantara Chapter 1', 'Kannada', 'Thriller', 330),
-> (408, 'Martin', 'Kannada', 'Action', 290),
-> (409, 'Sitaare Zameen Par', 'Hindi', 'Drama', 280),
-> (410, 'War 2', 'Hindi', 'Action', 380),
-> (411, 'Dragon', 'Tamil', 'Drama', 260),
-> (412, 'HIT 3', 'Telugu', 'Crime', 310),
-> (413, 'Tourist Family', 'Tamil', 'Family', 220),
-> (414, 'Retro', 'Tamil', 'Romance', 240),
-> (415, 'Thug Life', 'Tamil', 'Action', 360),
-> (416, 'Lucifer 2', 'Malayalam', 'Action', 330),
-> (417, 'Empuraan', 'Malayalam', 'Thriller', 350),
-> (418, 'Marco', 'Malayalam', 'Crime', 300),
-> (419, 'Deva', 'Hindi', 'Action', 290),
-> (420, 'Bhool Chuk Maaf', 'Hindi', 'Comedy', 240),
-> (421, 'Kanguva', 'Tamil', 'Fantasy', 370),
-> (422, 'Vidaamuyarchi', 'Tamil', 'Thriller', 320),
-> (423, 'Game Changer', 'Telugu', 'Political Drama', 340),
-> (424, 'Indian 3', 'Tamil', 'Action', 360),
-> (425, 'Baaghi 4', 'Hindi', 'Action', 310);
```

Windows taskbar at the bottom shows the Start button, a search bar, and various application icons including File Explorer, Edge, and WhatsApp. The system tray on the right displays the language as 'ENG IN', signal strength, volume, and a battery level of 100% at 12:57 on 15-07-2026.

```
MySQL 8.0 Command Line C... + v
mysql> CREATE TABLE bookings (
-> booking_id INT PRIMARY KEY,
-> customer_name VARCHAR(100) NOT NULL,
-> movie_id INT NOT NULL,
-> number_of_tickets INT NOT NULL,
-> booking_date DATE NOT NULL,
-> FOREIGN KEY (movie_id) REFERENCES movies(movie_id)
-> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO bookings (booking_id, customer_name, movie_id, number_of_tickets, booking_date) VALUES
-> (501, 'Aarav Gupta', 401, 2, '2026-03-01'),
-> (502, 'Siya Sharma', 402, 4, '2026-03-01'),
-> (503, 'Rohan Verma', 403, 3, '2026-03-02'),
-> (504, 'Ananya Singh', 404, 2, '2026-03-02'),
-> (505, 'Kabir Patel', 405, 5, '2026-03-03'),
-> (506, 'Ishita Joshi', 406, 2, '2026-03-03'),
-> (507, 'Vihaan Mehta', 407, 1, '2026-03-04'),
-> (508, 'Myra Nair', 408, 3, '2026-03-04'),
-> (509, 'Arjun Das', 409, 2, '2026-03-05'),
-> (510, 'Kiara Roy', 410, 4, '2026-03-05'),
-> (511, 'Aditya Kumar', 411, 2, '2026-03-06'),
-> (512, 'Aditi Shah', 412, 3, '2026-03-06'),
-> (513, 'Dev Yadav', 413, 4, '2026-03-07'),
-> (514, 'Riya Jain', 414, 2, '2026-03-07'),
-> (515, 'Yash Saini', 415, 5, '2026-03-08'),
-> (516, 'Mehak Khan', 416, 1, '2026-03-08'),
-> (517, 'Aryan Mishra', 417, 3, '2026-03-09'),
-> (518, 'Tanvi Agarwal', 418, 2, '2026-03-09'),
-> (519, 'Kunal Malhotra', 419, 4, '2026-03-10'),
-> (520, 'Pari Kaur', 420, 2, '2026-03-10');
Query OK, 20 rows affected (0.01 sec)
Records: 20 Duplicates: 0 Warnings: 0

mysql> UPDATE movies
-> SET ticket_price = 400
-> WHERE movie_name = 'War 2';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE movies
```

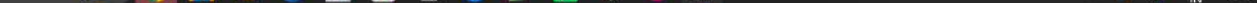
```
MySQL 8.0 Command Line CLI x + -
Query OK, 1 row affected (0.00 sec)

mysql>
mysql>
mysql> DELETE FROM movies
    -> WHERE movie_name = 'Dragon';
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key constraint fails ('online_sopping_system`.`bookings`, CONSTRAINT `bookings_ibfk_1` FOREIGN KEY (`movie_id`) REFERENCES `movies` (`movie_id`))
mysql>
mysql> DELETE FROM bookings
    -> WHERE customer_name = 'Kabir Patel'
    -> AND booking_date = '2026-03-03';
Query OK, 1 row affected (0.01 sec)

mysql> SELECT b.customer_name,
    -> m.movie_name
    -> FROM bookings b
    -> INNER JOIN movies m
    -> ON b.movie_id = m.movie_id;
+-----+-----+
| customer_name | movie_name |
+-----+-----+
| Aarav Gupta   | Pushpa 3   |
| Siya Sharma   | Kalki 2898 AD |
| Rohan Verma   | Chhaava    |
| Ananya Singh  | Housefull 5 |
| Ishita Joshi  | Jailer 2   |
| Vihaan Mehta  | Kantara Chapter 1 |
| Myra Nair     | Martin     |
| Arjun Das     | Sitaare Zameen Par |
| Kiara Roy     | War 2      |
| Aditya Kumar  | Dragon     |
| Aditi Shah    | HIT 3      |
| Dev Yadav     | Tourist Family |
| Riya Jain     | Retro      |
| Yash Saini    | Thug Life  |
| Mehak Khan    | Lucifer 2  |
| Aryan Mishra  | Empuraan   |
| Tanvi Agarwal | Marco      |
| Kunal Malhotra | Deva       |
| Pari Kaur     | Bhool Chuk Maaf |
+-----+-----+
```

customer_name	language	booking_date
Aarav Gupta	Telugu	2026-03-01
Siya Sharma	Telugu	2026-03-01
Rohan Verma	Hindi	2026-03-02
Ananya Singh	Hindi	2026-03-02
Ishita Joshi	Tamil	2026-03-03
Vihaan Mehta	Kannada	2026-03-03
Myra Nair	Kannada	2026-03-04
Arjun Das	Hindi	2026-03-05
Kiara Roy	Hindi	2026-03-05
Aditya Kumar	Tamil	2026-03-06
Aditi Shah	Telugu	2026-03-06
Dev Yadav	Tamil	2026-03-07
Riya Jain	Tamil	2026-03-07
Yash Saini	Tamil	2026-03-08
Mehak Khan	Malayalam	2026-03-10
Aryan Mishra	Malayalam	2026-03-09
Tanvi Agarwal	Malayalam	2026-03-09
Kunal Malhotra	Hindi	2026-03-10
Pari Kaur	Hindi	2026-03-10

```
mysql> SELECT b.customer_name,  
->         m.genre,  
->         b.number_of_tickets  
-> FROM bookings b  
-> INNER JOIN movies m  
-> ON b.movie_id = m.movie_id;
```



The taskbar shows the Start button, a search bar, and several pinned application icons including Google Chrome, File Explorer, Microsoft Edge, Spotify, OneDrive, a calendar, a mail client, a social media app, a game, a messaging app with 66 notifications, a clock, and a web browser. The system tray on the right displays the date and time as 12:57 on 15-07-2026, along with icons for network, volume, and battery status.

MySQL 8.0 Command Line CLI

19 rows in set (0.00 sec)

mysql> SELECT b.customer_name,
-> m.movie_name,
-> m.language
-> FROM bookings b
-> INNER JOIN movies m
-> ON b.movie_id = m.movie_id
-> WHERE m.language = 'Hindi';

customer_name	movie_name	language
Rohan Verma	Chhaava	Hindi
Ananya Singh	Housefull 5	Hindi
Arjun Das	Sitaare Zameen Par	Hindi
Kiara Roy	War 2	Hindi
Kunal Malhotra	Deva	Hindi
Pari Kaur	Bhool Chuk Maaf	Hindi

6 rows in set (0.00 sec)

mysql> SELECT b.customer_name,
-> m.movie_name,
-> m.language
-> FROM bookings b
-> INNER JOIN movies m
-> ON b.movie_id = m.movie_id
-> WHERE m.language = 'Hindi';

customer_name	movie_name	language
Rohan Verma	Chhaava	Hindi
Ananya Singh	Housefull 5	Hindi
Arjun Das	Sitaare Zameen Par	Hindi
Kiara Roy	War 2	Hindi
Kunal Malhotra	Deva	Hindi
Pari Kaur	Bhool Chuk Maaf	Hindi

6 rows in set (0.00 sec)

mysql>

Search

ENG IN 100% 12:58 15-07-2026